CALCULATING OPTIMUM BAND COMBINATION USING OIF INDEXING: A CASE STUDY ON CHINSURAH MUNICIPALITY, HOOGHLY, WEST BENGAL

1. INTRODUCTION:

Remote sensing is an excellent integration of science and technology. It promises to ensure sustainable use of resources; natural, economy and human. The growing number of earth observation satellites and the continuing improvements of their remote sensing systems provide mankind with unprecedented global capacity for systematic monitoring of the earth surface. Large amount or remote sensing data is generated every day but only a small fraction is used for production of information. The remote sensing sensors or instruments used for the purpose of earth observations record information in a definite range of electromagnetic spectrum of the radiation. The determination of the optimal combination of spectral intervals providing the maximum information with the minimum number of intervals is of principal importance.

OIF (Optimum Index Factor) OIF is a statistical calculation of every possible three-bands rendered as R-G-B. It is developed by Chavez et al. Values of OIF were determined in order to designate the most favourable band combination and rank the bands according to their statistical information. It is based on the amount of total variance and correlation between various bands.

2. STUDY AREA:

Chinsurah city is the administrative headquarter of Hooghly district in West Bengal. It is situated between 22° 56’ 19’’ and 22° 52’ 7’’ north latitude and between 88° 22' 1’’ and 88° 24’ 14’’ east longitude. It has an area of 22.2927 square kilometers a total population of 177,833 (2011).

It is situated by the banks of Hoogly river. On the north and north-east the district is bounded by Bandel and Bansberia. On the south it is bounded by Chandannagar and on the west by Akna and Kodalia. The areas to the east and south-east are low-lying alluvial plains, that

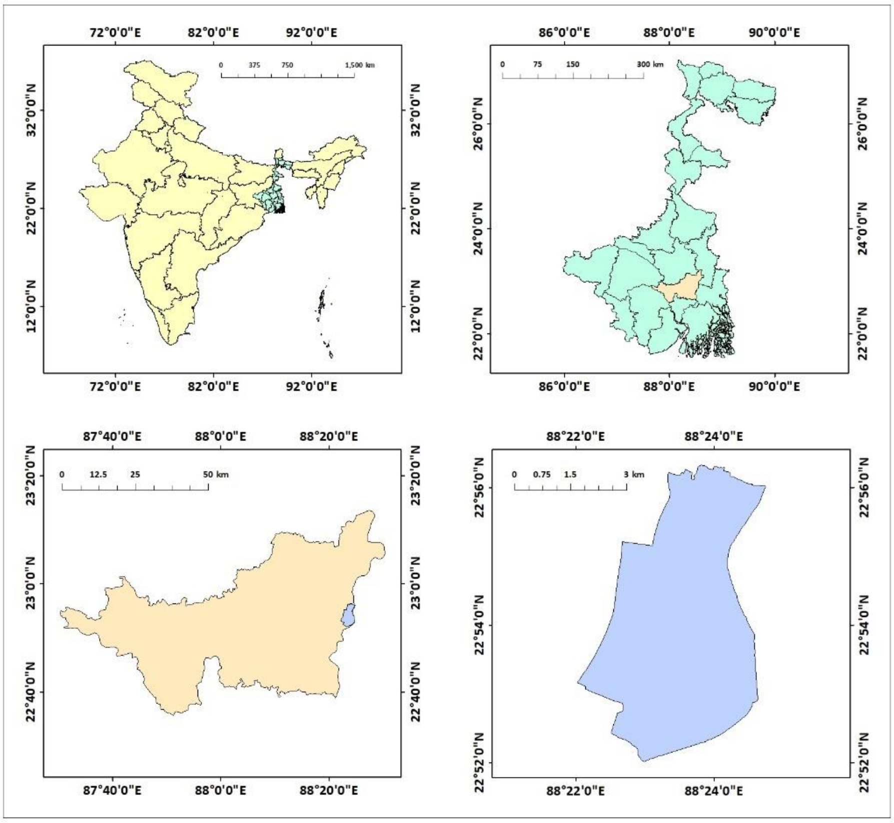


Figure 1: Location map of Chinsurah

forms part of the Gangetic delta The high west bank of the tidal Hooghly river is highly industrialised.

3. DATA SOURCES AND METHODOLOGY:

DATA SET	TAKEN ON	SOURCE
Landsat OLI Data	09/11/2000	https://earthexplorer.usgs.gov
Landsat TM Data	19/11/2021	https://earthexplorer.usgs.gov

The data set in this study consists of: Landsat OLI data; Acquisition Date: November 09, 2000.

Landsat TM data; Acquisition Date: November 19, 2021.

For the calculation of the co-relation statistics the tool named "band collection statistics' of ArcGis software have been used. Then, the matrixes are generated by using the data previously generated from the above-mentioned method. The calculations have been done by using the following formulas-

OIF = ∑SD / ∑r

(Where, ∑SD- sum of all the three bands in within a combination

∑r- sum of co-relation between the three bands)

4.RESULT AND DISCUSSION:

TABLE 1: The standard deviation and co-relation matrixes are given below -

For Landsat OLI Data-

BAND NO.	2	3	4	5	6	7
SD	388.8234	465.9584	703.7992	2296.863	1995.688	1408.332

For Landsat TM Data-

BAND NO.	1	2	3	4	5	7
SD	4.4729	2.8273	4.6287	11.3638	18.0512	10.0465

TABLE 2: Correlation matrix –

For Landsat OLI Data-

BAND NO.	**2**	**3**	**4**	**5**	**6**	**7**
2	1	0.9246	0.86297	-0.22339	0.26272	0.43251
3	0.9246	1	0.94326	-0.17884	0.2727	0.43193
4	0.86297	0.94326	1	-0.19416	0.38977	0.5642
5	-0.22339	-0.17884	-0.19416	1	0.64297	0.41579
6	0.26272	0.2727	0.38977	0.64297	1	0.94813
7	0.43251	0.43193	0.5642	0.41579	0.94813	1

For Landsat TM Data-

BAND NO.	**1**	**2**	**3**	**4**	**5**	**7**
1	1	0.86663	0.84274	-0.25451	0.27159	0.45545
2	0.86663	1	0.92947	-0.12184	0.32942	0.49532
3	0.84274	0.92947	1	-0.15602	0.4144	0.60818
4	-0.25451	-0.12184	-0.15602	1	0.60137	0.32727
5	0.27159	0.32942	0.4144	0.60137	1	0.91538
7	0.45545	0.49532	0.60818	0.32727	0.91538	1

From the above mentioned 2 charts the OIF values have been calculated. In total 20 band combinations were generated from per-mutation of Landsat 8 (2, 3, 4, 5, 6, 7) and Landsat 5 (1, 2, 3, 4, 5, 7) except the thermal band and the calculations are done accordingly. Table 3 shows the calculated OIF and values for different groups (VISIBLE, NIR, SWIR-1, SWIR-2). Arrangement of bands in any band colour combination as R-G-B doesn't affect on the values of OIF, it will be the same.

TABLE 3: Calculation of OIF values-

For Landsat OLI Data-

SL NO.	BAND	OIF	SL NO.	BAND	OIF
1	**234**	570.7353	11	**345**	6079.017
2	**235**	6033.356	12	**346**	1971.344
3	**236**	1952.35	13	**347**	1329.33
4	**237**	1264.988	14	**356**	6458.083
5	**245**	**7609.639**	15	**357**	6236.026
6	**246**	2037.87	16	**367**	2341.525
7	**247**	1344.831	17	**456**	5958.108
8	**256**	6861.167	18	**457**	5610.621
9	**257**	6551.373	19	**467**	2159.623
10	**267**	2307.981	20	**576**	2840.656

For Landsat TM data-

SL NO.	BAND	OIF	SL NO.	BAND	OIF
1	**123**	4.520509	11	**234**	28.882
2	**124**	38.06804	12	**235**	15.24374
3	**125**	17.27358	13	**237**	8.609325
4	**127**	9.544789	14	**245**	39.85698
5	**134**	47.35059	15	**247**	34.58808
6	**135**	17.76167	16	**257**	18.38268
7	**137**	10.04427	17	**345**	39.59721
8	**145**	**54.79489**	18	**347**	33.40775
9	**147**	49.00172	19	**357**	16.88704

10	**157**	19.83086	20	**457**	21.39971

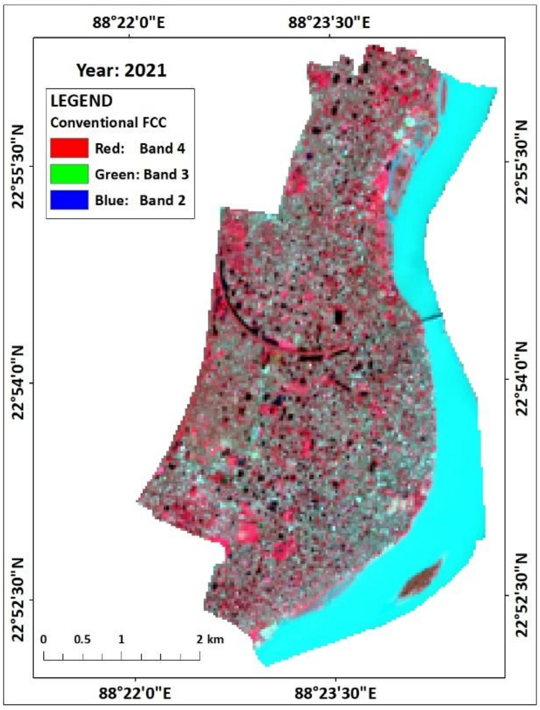


Figure 2: Conventional FCC

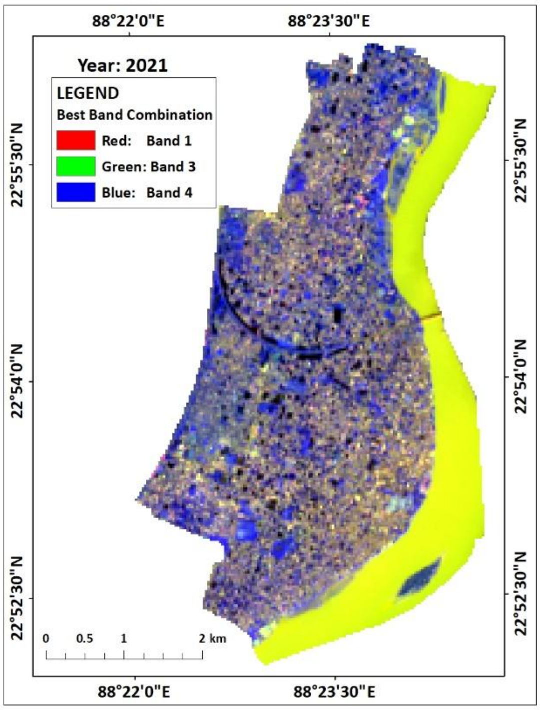
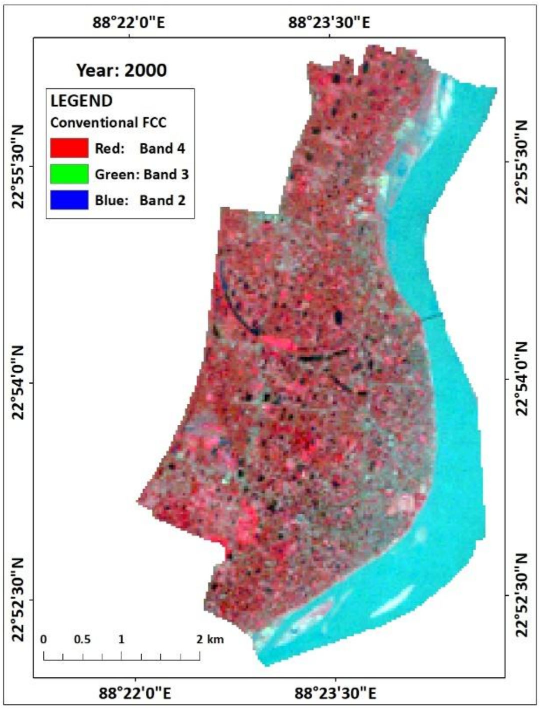


Figure3: Best band combination for Landsat OLI



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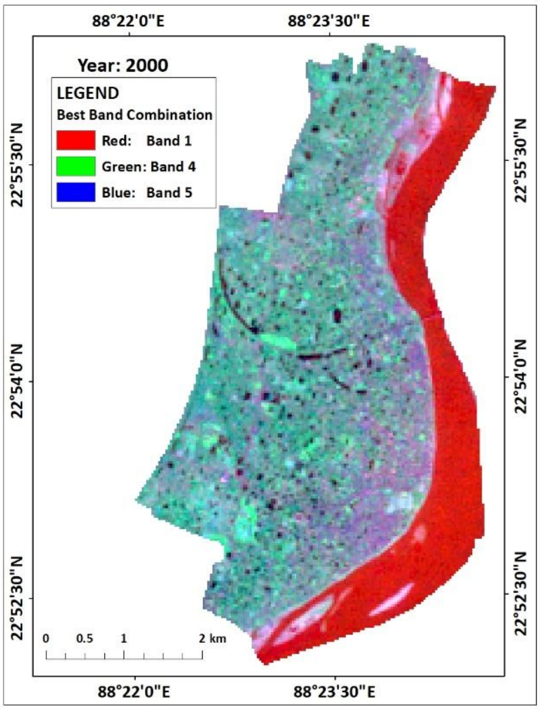


Figure5: Best band combination for Landsat TM

According to the calculation the highest OIF value is 7609.639 for the band combination **2-4-5 (BLUE-RED-NIR)** for Landsat OLI Data. The highest OIF value is 54.79489 for the band combination **1-4-5 (BLUE-NIR-SWIR)** for Landsat TM Data.

5. CONCLUSION:

Using of OIF and 3D Indexing technique for evaluating the bands of OLI and TM data simplified the selection of suitable bands for discriminating the different LULC. It also reduces the time of selection. Application of the technique to the selected bands of OLI and TM data resulted in 20 combinations for each data set. Based on the results obtained from the calculations according to OIF method for OLI data 2-4-5(BLUE-RED-NIR) and for TM data 1-4-5(BLUE-NIR-SWIR) are the best combinations.

TEMPORAL BASED LANDUSE LANDCOVER ANALYSIS: A CASE STUDY ON LANDUSE COVER CHANGING AREAL STATISTICS AND LAND USE MIX

1. INTRODUCTION:

The surface of the Earth includes a variety of natural and artificial geographical features such as ecosystems, landforms, human settlements, and engineered constructions. Land use and land cover (LULC) analysis is a general term used to depict Earth surface cover, whether it is natural or manmade. In the past three decades, the cities in developing countries have experienced a rapid increase in the rate of the population growth. According to the world urbanization prospects 2014 revision, the trend in urbanization recently has been focusing on the population transition between urban and rural. Developing land for an urbanization and migration of people from rural areas is a global phenomenon.

The expanding network of roads and increasing reliance on the automobile, population began shifting from cities to fringe and thus the land use pattern of those adjacent areas changed very rapidly. The spatial patterns of urban growth over different time periods can be systematically mapped, monitored and accurately assessed from satellite data (remotely sensed data) along with conventional ground data. The recent technologies like GIS and remote sensing helps in identifying the pattern of growth and its rate. Hence the present attempt has been made at micro level by selecting Chinsurah urban area of the district of Hooghly, West Bengal. This report is an attempt of understanding the impact and characteristics of LULC elements and its affect on the other land use land cover.

2. STUDY AREA:

Chinsurah city is the administrative headquarter of Hooghly district in West Bengal. It is situated between 22° 56’ 19’’ and 22° 52’ 7’’ north latitude and between 88° 22' 1’’ and 88° 24’ 14’’ east longitude. It has an area of 22.2927 square kilometers a total population of 177,833 (2011).

It is situated by the banks of Hooghly river. On the north and north-east the district is bounded by Bandel and Bansberia. On the south it is bounded by Chandannagar and on the west by Akna and Kodalia. The areas to

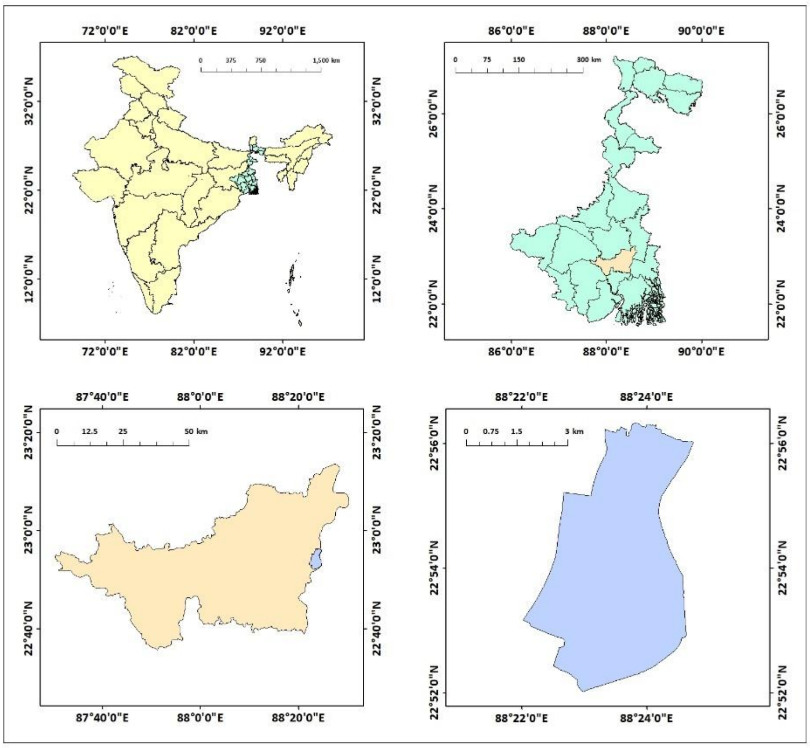


Figure 6: Location map of Chinsurah

the east and south-east are low-lying alluvial plains, that forms part of the Gangetic delta The high west bank of the tidal Hooghly river is highly industrialised.

3. DATA SOURCES AND METHODOLOGY:

Multi-temporal satellite data are used for this study. Landsat Thematic Mapper (TM-5) images of **2000** and **2007** and Operational Land Imager-8 (OLI-8) image for 2014 and 2021 are used for measurement of temporal urban growth pattern. The Landsat multi-temporal data are collected from the United States Geology Survey (https://earthexplorer.usgs.gov/).

SATELLITE	SENSOR	DATE OF AQUISITION	SPATIAL RESOLUTION	RADIOMETRIC RESOLUTION	PATH	ROW
Landsat_5	TM	09/11/2000	30m	8 bit	138	44
Landsat_5	TM	12/11/2007	30m	8 bit	138	44
Landsat_8	OLI	16/11/2014	30m	16 bit	138	44
Landsat_8	OLI	19/11/2021	30m	16 bit	138	44

The boundary map and shapefile of Chinsurah town were obtained through geo referencing techniques. The collected images were pre-georeferenced to the UTM-Zone 45 North Projection with WGS-84 datum (Path/Row: 138/44). Based on these data image classification, accuracy assessment, transition matrix, land use mix etc. are done.

4. RESULT AND DISCUSSION:

4.1 TEMPORAL IMAGE CLASSIFICATION OF CHINSURAH:

Image Classification is a fundamental task that attempts to comprehend an entire image as a whole. The goal is to classify the image by assigning it to a specific label. Typically, Image Classification refers to images in which only one object appears and is analyzed. In contrast, object detection involves both classification and localization tasks, and is used to analyze more realistic cases in which multiple objects may exist in an image.

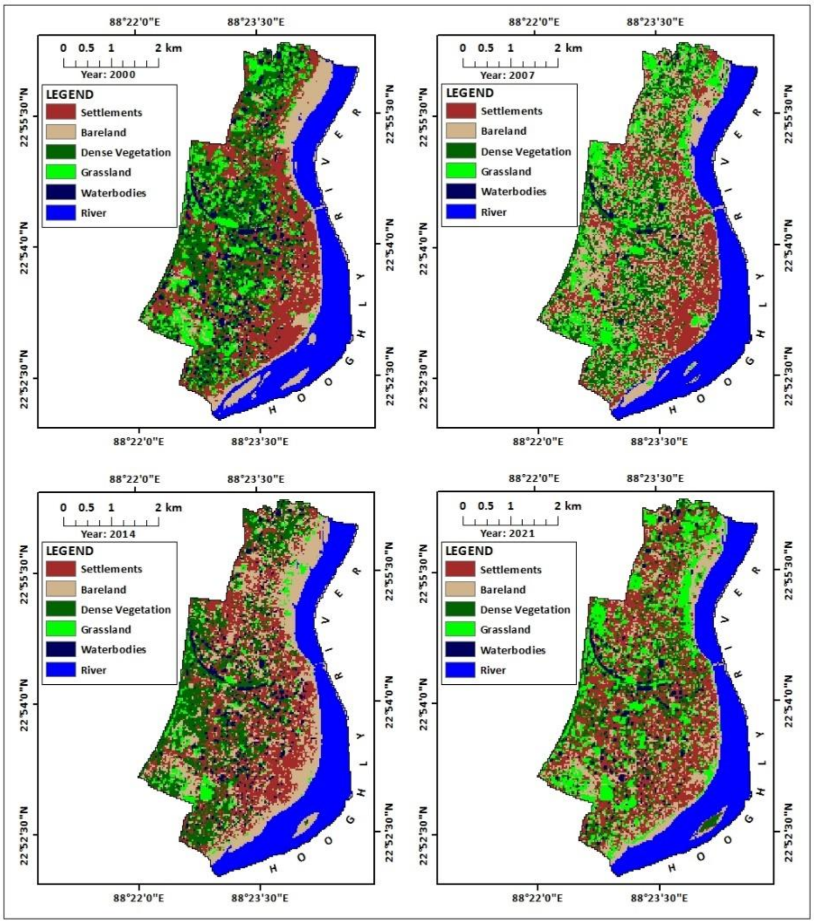


Figure 7: LULC Classification of Chinsurah

Here Supervised Classification by using Maximum Likelihood Method through ERDAS Imagine software have been done. Supervised

Classification is a method where the users take the training samples from the satellite image and on the basis of that samples the signature file gets generated. Through that signature file the system generates the classification image.

Here 4 years data have been chosen which are **2000,2007,2014,2021.** In these classified images six classes have been generated which are respectively Settlements, Bare land, Grassland, Dense Vegetation, Waterbodies, River.

By the above images we can observe that the Settlement areas are growing faster and Dense Vegetation areas are decreasing faster in comparison to the other four classes. The area of Bare land and Grassland are the most unstable portions and the Waterbodies and River areas are almost constant. All the areas are given bellow.

LAND USE	AREA_2000 (sq.km.)	AREA_2007 (sq.km.)	AREA_2014 (sq.km.)	AREA_2021 (sq.km.)
SETTLEMENT	5.772600	4.424399	5.545800	6.885000
BARELAND	2.868299	5.837400	5.636699	3.596399
DENSE VEGETATION	4.622400	4.593600	4.621500	2.701800
GRASSLAND	3.122999	2.754900	1.031400	3.895199
WATERBODIES	1.740599	0.463500	1.159199	0.977399
RIVER	4.520699	4.573799	4.652999	4.591800

4.2 ACCURACY ASSESSMENT OF CLASSIFIED IMAGES:

4.2.1 Using Contingency Method:

There are many different ways to look at the thematic accuracy of a classification. The error matrix allows us to calculate the following accuracy metrics.

- User Accuracy
- Commission Error
- Producer Accuracy
- Omission Error
- Overall Accuracy
- Accuracy statistics (e.g. kappa co-efficient)

For the Accuracy Assessment of the classified images from contingency matrix which have been generated by ERDAS Imagine Software. The columns always represent the reference data or the data that is known to be true or corrects. The rows are the mapped classes that were generated from the remote sensed data. The error matrix allows us to be able to calculate a variety of accuracy metrics from our data.

Table 1: Accuracy assessment of 2000 Land use image-

Data	Waterbodies	River	Dense Vege	Grassland	Settlement	Bareland	Row Total	G	User Accuracy	Comission Error
Waterbodies	153	0	2	0	0	0	155	23870	0.987096774	0.012903226
River	0	2989	0	0	0	4	2993	9104706	0.998663548	0.001336452
Dense Vege	0	0	325	5	0	0	330	108240	0.984848485	0.015151515
Grassland	0	0	1	113	0	2	116	14036	0.974137931	0.025862069
Settlement	0	0	0	0	454	1	455	206570	0.997802198	0.002197802
Bareland	1	53	0	3	0	232	289	69071	0.802768166	0.197231834
Column Total	154	3042	328	121	454	239	4338	9526493		
Producer Accuracy	0.993506494	0.982577252	0.9908537	0.933884298	1	0.970711297				
Omission Error	0.006493506	0.017422748	0.0091463	0.066115702	0	0.029288703				

Class	Waterbodie	River	Dense Vege	Grassland	Settlement	Bareland	Total
Value	153	2989	325	113	454	232	4266
						Overall Accuracy	98.34024896

C	C*T	G	(C*T)-G	T^2	(T^2)-G	(C*T)-G/ (T^2)-G	Kappa Statistics	Strenght Of Agreement
4266	494856	9526493	-9031637	13456	-9513037	0.949395761	0.81-0.99	Almost perfect

Table 2: Accuracy assessment of 2007 Land use image-

Data	Waterbodies	River	Dense Vege	Grassland	Bareland	Settlement	Row Total	G	User Accuracy	Comission Error
Waterbodies	86	7	2	0	0	0	95	8265	0.905263158	0.094736842
River	1	2999	0	0	0	0	3000	9129000	0.999666667	0.000333333
Dense Vege	0	0	115	34	2	0	151	20687	0.761589404	0.238410596
Grassland	0	0	13	175	1	0	189	39879	0.925925926	0.074074074
Bareland	0	37	7	2	202	2	250	60000	0.808	0.192
Settlement	0	0	0	0	35	70	105	7560	0.666666667	0.333333333
Column Total	87	3043	137	211	240	72	3790	9265391		
Producer Accuracy	0.988505747	0.985540585	0.8394161	0.829383886	0.841667	0.972222222				
Omission Error	0.011494253	0.014459415	0.1605839	0.170616114	0.158333	0.027777778				

Class	Waterbodie	River	Dense Vege	Grassland	Bareland	Settlement	Total
Value	86	2999	115	175	202	70	3647
						Overall Accuracy	96.22691293

C	C*T	G	(C*T)-G	T^2	(T^2)-G	(C*T)-G/ (T^2)-G	Kappa Statistics	Strenght Of Agreement
3647	346465	9265391	-8918926	9025	-9256366	0.963545089	0.81-0.99	Almost perfect

Table 3: Accuracy assessment of 2014 Land use image-

Data	Waterbodies	River	Dense Vege	Grassland	Bareland	Settlement	Row Total	G	User Accuracy	Comission Error
Waterbodies	144	0	4	0	0	0	148	23384	0.972972973	0.027027027
River	0	2638	0	0	0	0	2638	7059288	1	0
Dense Vege	11	0	29	1	0	1	42	1386	0.69047619	0.30952381
Grassland	0	0	0	64	2	0	66	4290	0.96969697	0.03030303
Bareland	0	38	0	0	231	2	271	63143	0.852398524	0.147601476
Settlement	3	0	0	0	0	234	237	56169	0.987341772	0.012658228
Column Total	158	2676	33	65	233	237	3402	7207660		
Producer Accuracy	0.911392405	0.985799701	0.878787879	0.984615385	0.991416	0.987341772				
Omission Error	0.088607595	0.014200299	0.121212121	0.015384615	0.008584	0.012658228				

Class	Waterbodie	River	Dense Vege	Grassland	Bareland	Settlement	Total
Value	144	2638	29	64	231	234	3340
						Overall Accuracy	98.17754262

C	C*T	G	(C*T)-G	T^2	(T^2)-G	(C*T)-G/ (T^2)-G	Kappa Statistics	Strenght Of Agreement
3340	140280	7207660	-7067380	1764	-7205896	0.980777408	0.81-0.99	Almost perfect

Table 4: Accuracy assessment of 2021 Land use image-

Data	Waterbodies	Dense Vege	Grassland	Settlement	Bareland	River	Row Total	G	User Accuracy	Comission Error
Waterbodies	101	7	0	0	0	0	108	11556	0.935185185	0.064814815
Dense Vege	6	93	3	24	1	0	127	13462	0.732283465	0.267716535
Grassland	0	1	260	0	11	0	272	75616	0.955882353	0.044117647
Settlement	0	5	2	373	19	0	399	162393	0.934837093	0.065162907
Bareland	0	0	13	10	273	93	389	118645	0.701799486	0.298200514
River	0	0	0	0	1	3563	3564	13029984	0.999719416	0.000280584
Column Total	107	106	278	407	305	3656	4859	13411656		
Producer Accuracy	0.943925234	0.877358491	0.9352518	0.916461916	0.895081967	0.974562363				
Omission Error	0.056074766	0.122641509	0.0647482	0.083538084	0.104918033	0.025437637				

Class	Waterbodies	Dense Vege	Grassland	Settlement	Bareland	River	Total
Value	101	93	260	373	273	3563	4663
						Overall Accuracy	95.9662482

C	C*T	G	(C*T)-G	T^2	(T^2)-G	(C*T)-G/ (T^2)-G	Kappa Statistics	Strenght Of Agreement
4663	503604	13411656	-12908052	11664	-13399992	0.963288038	0.81-0.99	Almost perfect

4.2.2 Using Signature Ellipse:

Signature ellipse plotting is a well-known technique to identify whether the classification of landcover maps is accurate or not. If the ellipse of different classes overlaps with each other then the classification may not be accurate.

ERDAS IMAGINE software has been used to create the signature ellipses for the classification maps of the four different years – **2000, 2007, 2014, 2021**.

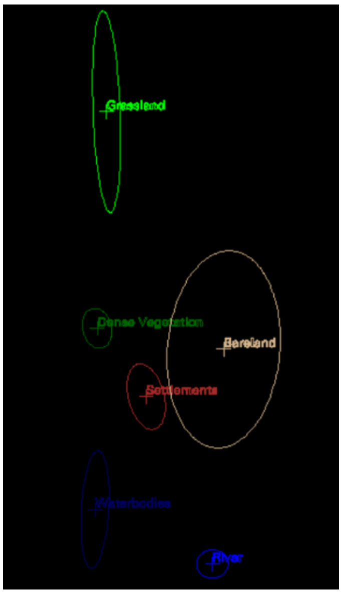


Figure 8: Signature Ellipse of LULC 2000

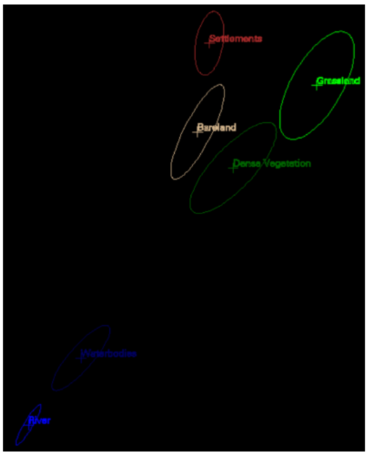


Figure 9: Signature Ellipse of LULC 2007

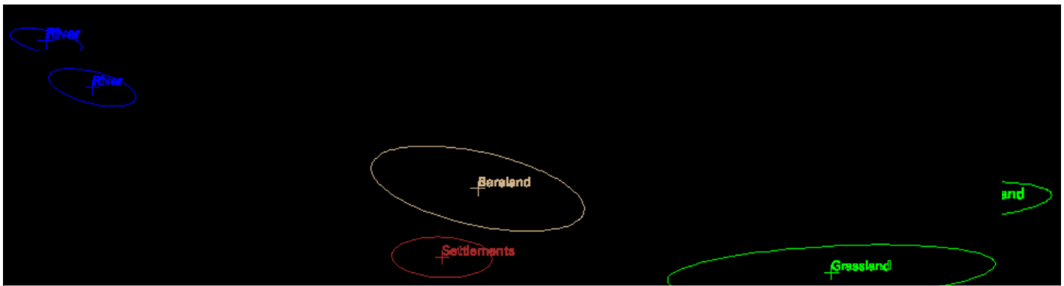


Figure 10: Signature Ellipse of LULC 2014

Figure 11: Signature Ellipse of LULC 2000

4.3 CALCULATION OF CHANGE INDEX:

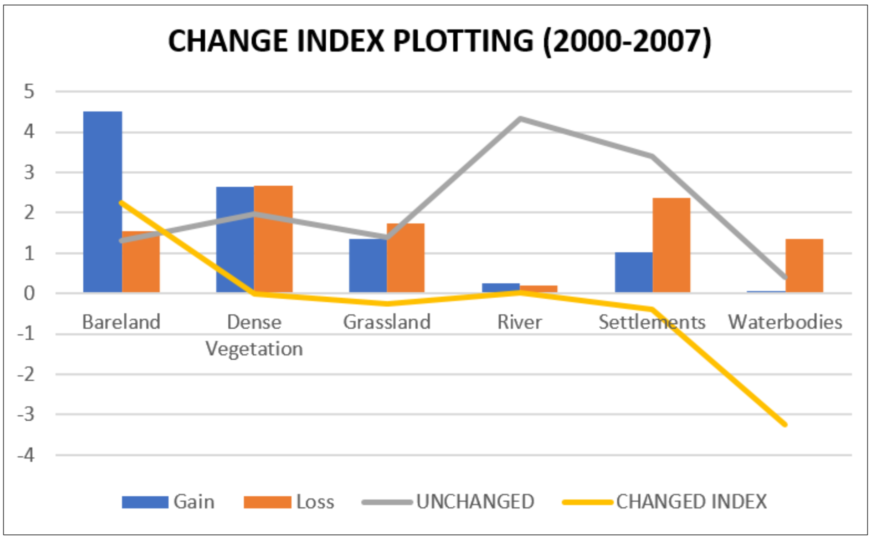
Through the change index we calculate that with in a time span how much area is gained and lost by a particular class. From this gained and lost area we can calculate the actual changed portion of that particular class. The formula for calculation of change index is-

$$\text{Change Index} = \frac{\text{Gained Area} - \text{Lost Area}}{\text{Unchanged Area}}$$

In time span between **2000-2007**, the lowest value can be seen in Waterbodies (-3.2471) and the highest value is Bare land (2.2580). The rest of the classes are lying in between -0.3 to 0.01. So from that we can say that the most stable class is Waterbodies land and the Unstable class is Bare land and the others are moderately stable.

Table 1: Change Index between 2000-2007-

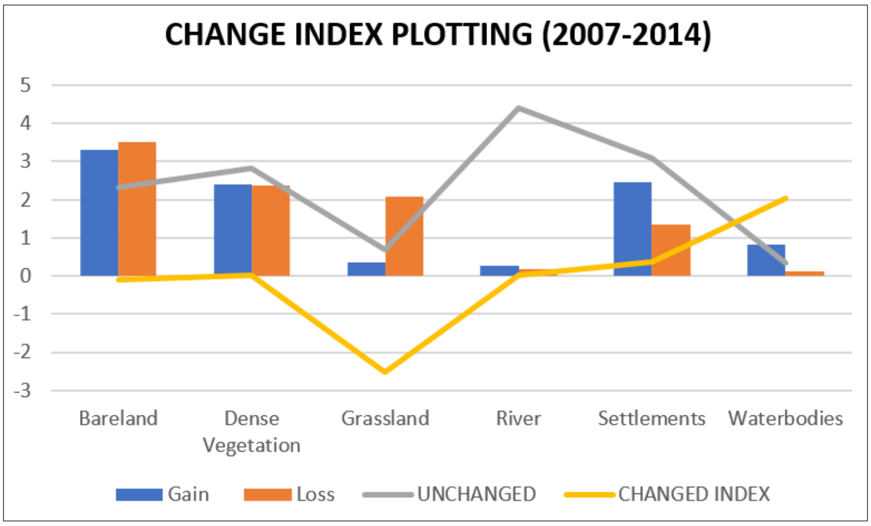
LAND USE	GAIN	LOSS	UNCHANGED	CHANGED INDEX
BARELAND	4.52250	1.55340	1.3149	2.258042437
DENSE VEGETATION	2.63070	2.65950	1.9629	-0.014672169
GRASSLAND	1.35540	1.72350	1.3995	-0.263022508
RIVER	0.24480	0.19170	4.329	0.012266112
SETTLEMENTS	1.02060	2.36880	3.4038	-0.396086727
WATERBODIES	0.07020	1.34730	0.3933	-3.247139588



In time span between **2007-2014**, the lowest value can be seen in Grassland (-2.5263) and the highest value is Waterbodies (2.0449). The rest of the classes are lying in between -0.08 to 0.36. So from that we can say that the most stable class is Grasslands and the Unstable class is Waterbodies and the others are moderately stable.

Table 2: Change Index between 2007-2014-

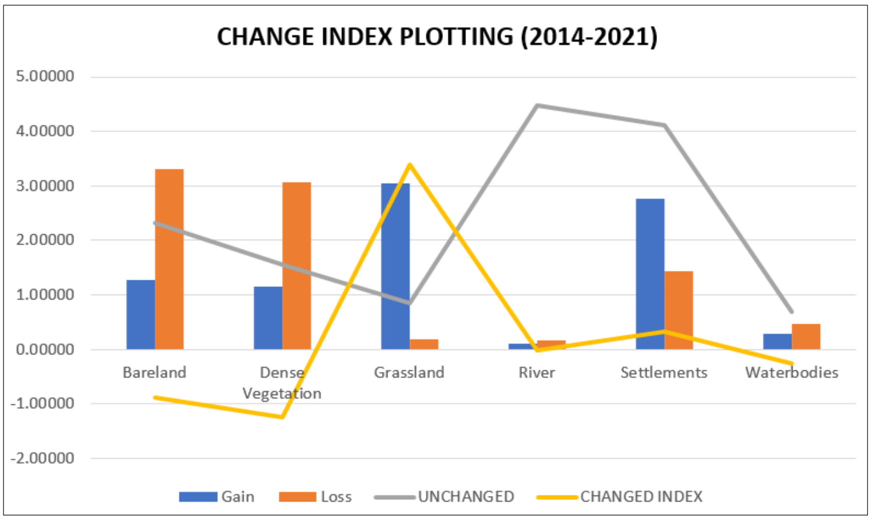
LAND USE	GAIN	LOSS	UNCHANGED	CHANGED INDEX
BARELAND	3.30840	3.50910	2.32830	-0.086200232
DENSE VEGETATION	2.40300	2.37510	2.21850	0.012576065
GRASSLAND	0.34920	2.07270	0.68220	**-2.526385224**
RIVER	0.26190	0.18270	4.39110	0.018036483
SETTLEMENTS	2.46960	1.34820	3.07620	0.364540667
WATERBODIES	0.81900	0.12330	0.34020	**2.044973545**



In time span between **2014-2021**, the lowest value can be seen in Dense vegetation (-1.2365) and the highest value is Grassland (3.3815). The rest of the classes are lying in between -0.87 to 0.32. So from that we can say that the most stable class is Dense vegetation and the Unstable class is Grassland and the others are moderately stable.

Table 3: Change Index between 2014-2021-

LAND USE	GAIN	LOSS	UNCHANGED	CHANGED INDEX
BARELAND	1.27350	3.31380	2.32290	-0.878341728
DENSE VEGETATION	1.14930	3.06900	1.55250	**-1.236521739**
GRASSLAND	3.04830	0.18450	0.84690	**3.381509033**
RIVER	0.11250	0.17370	4.47930	-0.013662849
SETTLEMENTS	2.76750	1.42830	4.11750	0.325245902
WATERBODIES	0.29070	0.47250	0.68670	-0.26474443



4.4 LANDUSE MIX CALCULATION:

Land use mix is a very important approach for taking different decisions of urban planning, rural planning etc. It is generally calculated in a grid manner. By calculating the information about variability of the data in within a unit area it can be obtained. Over here, for calculating the variability Entropy has been used so that the variability can be calculated in regard to the most dominant one.

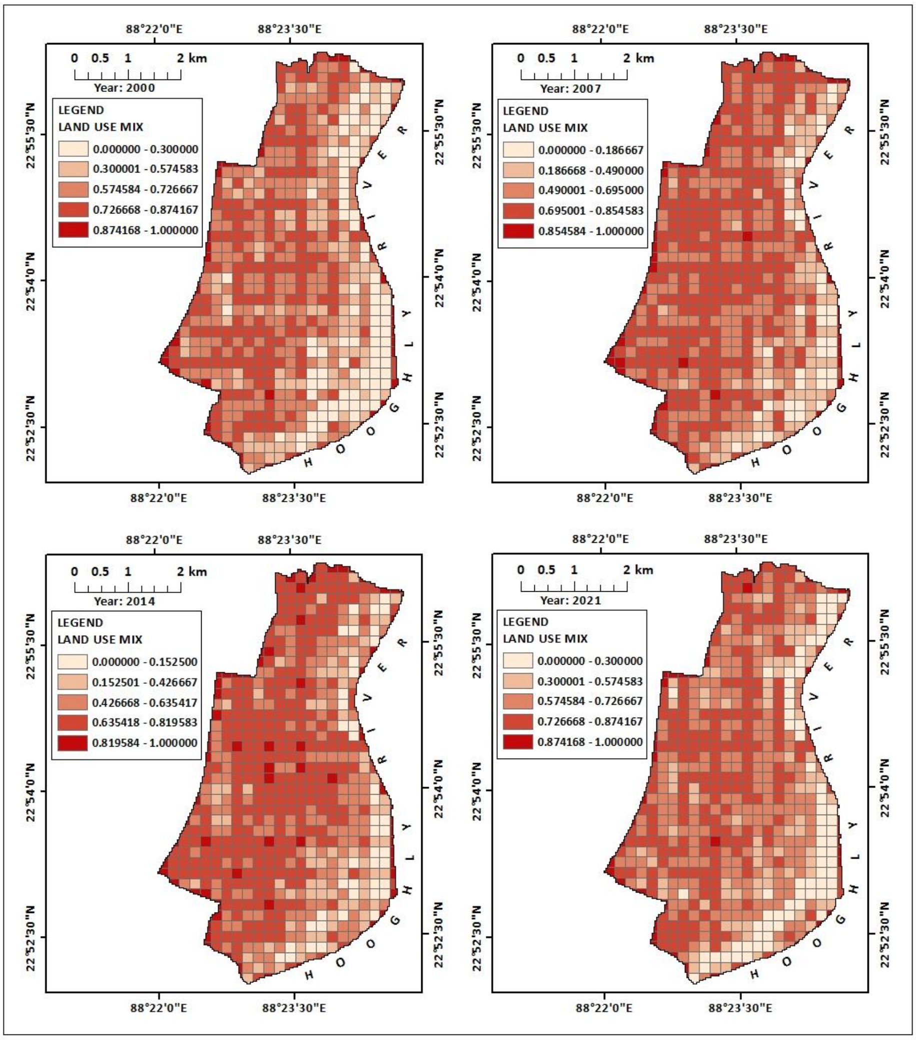


Figure 12: Maps of Land Use Mix

In the year **2000**, the highest mixing is located in the western and some middle regions (0.87-1.00) whereas the lowest mixing (0.00-0.30) can be located in the entire eastern, south eastern, north eastern and very few parts of western area.

In the year **2007**, the highest mixing is located in the few parts of western and south western regions (0.85-1.00) whereas the lowest mixing (0.00-0.18) can be located in the entire eastern, south-eastern, north-eastern area.

In the year **2014**, the highest mixing is located in the western and few parts of middle regions (0.81-1.00) whereas the lowest mixing (0.00-0.15) can be located in the entire eastern, south eastern, north eastern area.

In the year **2021**, the highest mixing is located in the south western regions (0.87-1.00) whereas the lowest mixing (0.00-0.30) can be located in the entire eastern, south eastern, north eastern and very few parts of western and middle areas.

5.CONCLUSION:

This report demonstrates Landuse - Landcover analysis and changing of chinsurah city of Hooghly and identifies the temporal and spatial changing patterns of elements by using multi temporal remote sensing images and GIS tool. Various analysts have been considered in quantifying the LULC changing pattern. All these studies have come up with different methodologies in quantifying landcover changes.

It has been found from the study that, settlements have shown a high population growth rate with the increasing no. of it. The urban area size has increased and vegetation area size has decreased the most during 2000-2021.